Experiment / replicate: _____

Experiment and starting materials

Use this worksheet to plan and record a run with the versioned CLASP-seq laboratory protocol. Copy the protocol's recipes, loading calculations and acceptance criteria into the appropriate laboratory records; this planning worksheet does not supply them.

Field	Your record
Experiment, replicate and batch identifiers	
Operator, laboratory and assay date	
Protocol version and link	
Starting point: sequence lists, plasmids or qualified cell libraries	
TCR library identifier and construct-manifest version	
HLA-peptide library identifier and construct-manifest version	
Cell derivatives, source and passage history	
Expression, recovery and representation records	
Reagent and instrument records	
Deviations from the protocol	

Experiment / replicate: _____

Sample map

The labels below are examples of roles. Replace them with your run identifiers. Add separate entries for additional controls and replicates.

Biological sample	Example 10x reaction label	Targeted PCR products	Actual sample/channel/index identifiers
Input receptor cells	IN_TCR	Input TCR	
Sorted experimental preparation	EXP_PAIRS	Experimental TCR and experimental HLA-peptide	
Input target cells	IN_HLA	Input HLA-peptide	

Both experimental PCR products derive from the same capture reaction. The four targeted products are pooled for Nanopore library preparation. The input reactions have independent barcode namespaces; identical barcode strings across reactions do not identify an experimental pair. Custom sample indexes identify amplicon samples and are separate from partition barcodes.

Experiment / replicate: _____

Controls and progression checks

Record the exact reference materials and acceptance criteria specified by the released protocol. Input-library samples characterize representation; pair-background assessment needs its own controls and analysis.

Check	Reference/control and criteria	Observed result and supporting file	Decision/deviation
Construct identity and library representation			
Cell identity, expression and recovery			
Fluorescence controls and sorting gates			
Positive and specificity/background controls			
Post-sort composition and recovery			
Targeted amplification and process blanks			
Fragment identity, quantification and pooling			
Sequencing, assignment and replicate checks			

Experiment / replicate: _____

Assay-day handoffs

Reserve sorting, compatible 10x capture and reverse transcription together. Record actual times rather than relying only on the planned schedule.

Stage	Responsible person / instrument	Planned time	Actual start/end	Hold, transfer or recovery record
Prepare and reserve input aliquots				
Label and pair cells				
Stabilize the preparation				
Sort and recover				
10x capture and reverse transcription				
Target recovery and library QC				
Nanopore preparation and sequencing				

Experiment / replicate: _____

Counts and loading

Keep these quantities separate. Use the protocol-specific loading worksheet and record how the measured input was converted to the intended load.

Quantity	Sample and stage	Value	Units / measurement method	Supporting record
Individual cells				
Acquired sorter events				
Co-positive sorter events				
Collected events				
Recovered cells				
Intact pairs, if assessed				
Loaded material				
Recovered partitions				

Experiment / replicate:

Analysis and outputs

Analysis code version, environment and configuration:

Library/reference manifest version:

Raw signal, reads, sample sheet and checksum record:

Assigned, ambiguous and excluded identities, with reasons:

Control and replicate results:

Pair-evidence tables and figures:

Model predictions, if made, stored separately with model settings:

Dataset location, reuse terms and version:

Retain unobserved combinations, insufficient evidence and technical failures as distinct states. A co-positive event or shared partition alone does not establish specific recognition.

Protocol and resources: https://tcrchallenge.org/clasp-seq/